



**Verified Carbon
Standard**

IMPROVED AGRICULTURAL LAND MANAGEMENT GROUPED CARBON PROJECT IN INDIA



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1 PROJECT DETAILS

1.1 Summary Description of the Project

The project activity facilitates the transition of smallholder farming systems toward sustainable and climate-resilient agricultural practices. Core interventions include reduced or zero tillage, incorporation of cover crops, and the replacement of synthetic fertilizers with organic nutrient inputs. These practices are designed to reduce greenhouse gas (GHG) emissions—specifically carbon dioxide (CO₂), methane (CH₄), and nitrous oxide (N₂O)—while enhancing atmospheric carbon removal through increased Soil Organic Carbon (SOC) sequestration. Through improved residue management, elevated biomass inputs, and reduced soil disturbance, the project simultaneously supports climate change mitigation and the long-term enhancement of soil fertility.

Developed by Asvata, the initiative is structured as a grouped project under the VM0042 v2.0 methodology (Methodology for Improved Agricultural Land Management), with a total potential implementation area of 20,000 hectares. This grouped approach allows for the phased inclusion of additional project activity instances over time. Crop selection—including cotton and other regionally relevant species—is tailored to agro ecological conditions and local agronomic practices to optimize both mitigation outcomes and farmer co-benefits.

To date, the project has enrolled 1,499 smallholder farmers managing a cumulative area of 2,222 hectares in and around Nagpur, Maharashtra, for implementation of improved Agricultural Land Management (ALM) practices. Baseline assessments indicate that approximately 80% of participating farmers previously practiced monoculture cotton cultivation, often followed by extended fallow periods. The remaining 20% engaged in short-duration cereal or wheat cropping. Conventional practices prevalent across the baseline scenario included high-intensity tillage and synthetic fertilizer application, both of which contribute to GHG emissions and soil degradation.

The transition to regenerative practices under this project is expected to improve soil biophysical properties, enhance productivity, and strengthen resilience to climatic variability. Emission reductions and removals are rigorously quantified and monitored in accordance with the methodological framework of VM0042, ensuring environmental integrity and verifiability. The project targets a cumulative mitigation potential of approximately 1,951,250 tCO₂e over a 30-year crediting period.

During the first monitoring period, spanning June 7, 2022, to April 30, 2025, the project was operational across 5,000 hectares, resulting in verified GHG mitigation outcomes totaling 119,583 tCO₂e.

This project represents an integrated approach to sustainable land management, delivering quantifiable climate benefits while promoting livelihood resilience among smallholder communities in Maharashtra and Telangana. It contributes meaningfully to national and global climate targets by mainstreaming regenerative agriculture practices at scale.

1.2 Audit History

As this project is presently under development, this section will be updated during project validation to include the monitoring periods.

Audit type	Period	Program	Validation/verification body name	Number of years
Validation	07-06-2022	VCS	Validation/verification body name	30 years
Verification	07-06-2022 – 30-04-2025	VCS		2 years 10 months

1.3 Sectoral Scope and Project Type

Sectoral scope	Sectoral Scope 14, Agriculture Forestry and Other Land Use (AFOLU)
AFOLU project category ¹	Agricultural Land Management (ALM)
Project activity type	Improved Cropland Management (ICM)

1.4 Project Eligibility

1.4.1 General eligibility

The project activity is within the VCS Program's scope. The project reduces the greenhouse gases (GHGs) listed in the Kyoto Protocol, including CO₂, CH₄, and N₂O from agricultural residue management, as well as N₂O from fertiliser reduction. The project adheres to the Improved Land Management (VM0042 v2.1) Methodology. The agricultural land management (ALM) approach offers methods for calculating the removals and reductions of greenhouse gas (GHG) emissions (CO₂, CH₄, and N₂O) brought about by the implementation of improved ALM practices.

Project eligibility according to the scope listed in section 2.1.1 in VCS Standard v4.7

S.No	Eligibility	Description/Justification
1	The seven Kyoto Protocol greenhouse gases	The project reduces the greenhouse gases (GHGs) listed in the Kyoto Protocol, including CO ₂ , CH ₄ , and N ₂ O from agricultural residue management, as well as N ₂ O from fertilizer reduction.
2	Ozone-depleting substances (ODS)	The project activity does not involve ODS.

¹ See Appendix 1 of the VCS Standard

3	Project activities supported by a methodology approved under the VCS Program through the methodology development and review process.	The project activity is supported by VM0042 methodology approved under the VCS Program.
4	Project activities supported by a methodology approved under an approved GHG program, unless explicitly excluded (see the Verra website for exclusions)	The project activity does not belong to any category which is excluded by Verra.
5	Jurisdictional REDD+ programs and nested REDD+ projects as set out in the Jurisdictional and Nested REDD+ (JNR) Requirements	The project is not a Jurisdictional REDD+ or nested REDD+ project.

Furthermore, the project is suitable under the terms of the VCS Programme because it cannot be classified into any of the excluded project categories listed in Table 1 of the VCS Standard v4.7 section 2.1.3.

AFOLU projects are expected to start the pipeline listing procedure within three years from the project start date, according section 3.8.2 of VCS standard v4.7. The project start date is 7th June-2022, and the three-year period is until 7th June 2025 before the listing of the project document has been completed with VERRA.

The VM0042 Methodology is applicable to a wide range of project activities that, in comparison to the baseline scenario, improve SOC storage and/or decrease net emissions of CO₂, CH₄, and N₂O from ALM operations. Since the project activity involves increasing SOC storage and lowering net emissions of CO₂, CH₄, and N₂O, it complies with the methodology.

1.4.2 AFOLU project eligibility

- The Project is an Agricultural Land Management (ALM) project, which is an eligible AFOLU project category in line with Appendix 1, VCS Standard 4.7. The Project is an eligible ALM project as it includes Improved Cropland Management activities that reduce and sequester net GHG emissions of cropland systems.
- The Project focuses on increasing soil organic carbon (SOC) storage in annual cropping systems through the implementation of regenerative agricultural practices as well as reducing net emissions of carbon dioxide (CO₂), methane (CH₄) and nitrogen (N₂O) through activities which improve the efficiency of fertiliser use. The Project activities follow the eligibility criteria laid out by the VM0042 Improved Agricultural Land Management version 2.1 approved by Verra on 11 sept 2024 (hereafter referred to as “VM0042”)
- The area where the Project activities occur has not been cleared, converted or degraded of native ecosystems throughout the 10-years prior to the project start date and does not

occur on a wetland. This will be assessed using GIS technology for each field submitted for verification and proof of this will be presented in the monitoring reports. This is a voluntary Project, meaning that the emission reductions and sequestration would not have taken place in the absence of the Project.

1.4.3 Transfer project eligibility

The project activity is not a transfer project and not seeking CPAs registration. Hence this section is not applicable.

1.5 Project Design

Indicate if the project has been designed as:

- ☐ Single location or installation
- ☐ Multiple locations or project activity instances (but not a grouped project)
- ☒ Grouped project

Grouped Project Design

Grouped projects are structured in such a way that they allow for project expansion in the phases following subsequent project validation. Based on the assumption that an increasing number of project activity instances (fields) will be adopting a set of eligible ALM practices and get enrolled in this project as the project expands within the defined project boundary. This means that this project will include further project activity instances (fields) during the subsequent validation/ verification phases of the project Validation will be based upon initial project activity instances located within the project area. The eligibility criteria for the inclusion of new project instances will comply with VCS Standard v4.7, and the VM0042 methodology criteria listed below.

Eligibility Criteria

The project instances will be eligible for inclusion in this grouped project as they meet eligibility criteria set out in 3.6.16- 3.6.17 of the VCS Standard v4.7 as described in table 1 below:

S.No	Eligibility Criterion	New project activity instances should comply
1	Meet the applicability conditions set out in the methodology applied to the project	The new project instance should meet the applicability conditions set in VM0042 Improved Agricultural Land Management V2.1 methodology
2	Use the technologies or measures specified in the project description.	The new project instance should follow sustainable agricultural practices as defined in the project

		description which includes reduction in the use of synthetic fertilizer, reduced tillage, crop rotation and crop residue management.
3	Apply the technologies or measures in the same manner as specified in the project description.	The new project instance should follow the activities mentioned in Description of the Project Activity in section 1.12
4	Are subject to the baseline scenario determined in the project description for the specified project activity and geographic area.	The new project activity instances will be installed in an Indian state subject to the same baseline scenario determined in Section 3.4.
5	Have characteristics with respect to additionality that are consistent with the initial instances for the specified project activity and geographic area. For example, the new project activity instances have financial, technical and/or other parameters (such as the size/scale of the instances) consistent with the initial instances, or face the same investment, technological and/or other barriers as the initial instances.	Each new project activity instance will demonstrate additionality as per methodology and standard requirements defined during initial validation
6	Occur within one of the designated geographic areas specified in the project description.	Each project activity instance will be in the geographic boundary specified in section 1.13.
7	Conform with at least one complete set of eligibility criteria for the inclusion of new project activity instances. Partial conformance with multiple sets of eligibility criteria is insufficient.	Each new project activity instance will comply with the complete set of eligibility criteria as established in this document
8	Be included in the monitoring report with sufficient technical, financial, geographic, and other relevant information to demonstrate conformance with the applicable set of eligibility criteria and enable evidence gathering by the validation/verification body	Each new project activity instance will be included in the monitoring report with sufficient technical, geographic, and other relevant information to demonstrate conformance with the eligibility criteria and provide evidence to the validation/verification body (VVB).
9	Have evidence of project ownership, in respect of each project activity instance, held by the project proponent from the respective start date of each project activity instance (i.e., the date upon which the project activity instance began reducing or removing GHG emissions).	Each new project activity instance will demonstrate ownership by the PP from the prospective start date.

10	Have a start date that is the same as or later than the grouped project start date	Each new project activity instance will have a start date that is same or later than the grouped project start date, which is 7 th June 2022.
11	Only be eligible for crediting from the later of start date of the project activity instance or the start of the verification period in which they were added to the grouped project, through to the end of the total project crediting period.	Each new project activity instance will be eligible for crediting from the start date of the project activity instance throughout to the end of the project crediting period.
12	Not be or have been enrolled in another VCS project.	Each new project activity instance will not be previously enrolled in another VCS project
13	Adhere to the clustering and capacity limit requirements for multiple project activity instances set out in 3.6.8 – 3.6.9.	Each new project activity instance must adhere to the clustering and capacity limit requirements.

1.6 Project Proponent

Organization name	Asvata Climate Solutions Private Limited
Role in the project	Project Developer
Contact person	Navin Mathur
Title	Chief operating officer
Address	Dr Annie Besant Rd, near Old Passport Office, Hanuman Nagar, Worli, Mumbai, Maharashtra 400030, India
Telephone	+91 98670 15373
Email	navin.mathur@asvata.com

1.7 Other Entities Involved in the Project

Organization name	RPG Foundation
Contact person	Manasi Rane
Title	Program lead
Address	Dr Annie Besant Rd, near Old Passport Office, Hanuman Nagar, Worli, Mumbai, Maharashtra 400030, India

Telephone	+91 9960378453
Email	vikram@rpgf.in

1.8 Ownership

The lands utilized for implementing the VCS project activity are legally owned by small and marginal farmers. These lands are under the implementation control of the Project Proponent (PP) through an agreement signed between the smallholder farmers and the PP at the start date of the VCS project activity, and are expected to remain under such control throughout the crediting period. This agreement complies with the VCS Standard Version 4.7, paragraph 3.7.1, point 6.

Additionally, as the rightful owners of their respective farmlands, the farmers have the authority to grant RPGF overall control and responsibility for the Improved Agricultural Land Management Project, as the Project Proponent, without infringing upon their rights. Each participating farmer formally assigns their carbon rights to RPGF through a signed agreement, which are subsequently transferred to ASVATA. The revenue generated from the sale of carbon offsets will be distributed among stakeholders in accordance with the policies and decisions set by ASVATA's management.

Supporting evidence of project ownership, such as MOUs, agreement documents, and land ownership records, are held by the Project Proponent and can be provided to the Validation/Verification Body upon request.

1.9 Project Start Date

Project start date	07-June-2022
Justification	The project start date, as defined by the VCS standard, is the commencement of activities aimed at generating reductions or removals of GHG emissions. For this project, the start date is 07 June-2022, on which the first activity to initiate the project, i.e., farmer engagement took place.

1.10 Project Crediting Period

Crediting period	☐ Seven years, twice renewable ☐ Ten years, fixed
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	☒ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements) <i>(According to 3.9.3 of VCS Standard 4.7, which states “For all AFOLU projects other than such ALM projects described in 3.9.2, the initial project crediting period shall be a minimum of 20 years up to a maximum of 100 years, which may be renewed at most four times, with a total project crediting period not to exceed 100 years.” The current project has a crediting period of 30 years)</i>
Start and end date of first or fixed crediting period	07-June-2022 to 06-June-2052

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

☒ < 300,000 tCO₂e/year (project)

☐ ≥ 300,000 tCO₂e/year (large project)

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
07-June-2022 to 31-December-2022	8,750
01-January-2023 to 31-December-2023	35,000
01-January-2024 to 31-December-2024	52,500
01-January-2025 to 31-December-2025	70,000
01-January-2026 to 31-December-2026	70,000
01-January-2027 to 31-December-2027	70,000
01-January-2028 to 31-December-2028	70,000
01-January-2029 to 31-December-2029	70,000
01-January-2030 to 31-December-2030	70,000
01-January-2031 to 31-December-2031	70,000
01-January-2032 to 31-December-2032	70,000
01-January-2033 to 31-December-2033	70,000
01-January-2034 to 31-December-2034	70,000
01-January-2035 to 31-December-2035	70,000
01-January-2036 to 31-December-2036	70,000

01-January-2037 to 31-December-2037	70,000
01-January-2038 to 31-December-2038	70,000
01-January-2039 to 31-December-2039	70,000
01-January-2040 to 31-December-2040	70,000
01-January-2041 to 31-December-2041	70,000
01-January-2042 to 31-December-2042	70,000
01-January-2043 to 31-December-2043	70,000
01-January-2044 to 31-December-2044	70,000
01-January-2045 to 31-December-2045	70,000
01-January-2046 to 31-December-2046	70,000
01-January-2047 to 31-December-2047	70,000
01-January-2048 to 31-December-2048	70,000
01-January-2049 to 31-December-2049	70,000
01-January-2050 to 31-December-2050	70,000
01-January-2051 to 31-December-2051	70,000
01-January-YYYY to 06-June-2052	35,000
Total estimated ERRs during the first or fixed crediting period	19,51,250
Total number of years	30
Average annual ERRs	65,041.67

1.12 Description of the Project Activity

RPGF Group, actively promotes regenerative and sustainable farming practices among farmers. These practices include, but are not limited to, crop rotation, crop residue management, improved land and nutrient management techniques—such as balancing the application of organic and inorganic nitrogen fertilizers. The specific practices implemented under this initiative are outlined below:

A. Crop Rotation

- i. Under baseline conditions (non-project farmers), over 80% of cotton farmers engage in monocropping. Typically, after harvesting cotton in January or February, fields are left fallow for approximately nine months, from March through May.

- ii. In contrast, farmers participating in the IALM project utilize their land throughout the year by adopting cotton-wheat or cotton-bengal gram cropping sequences. This ensures continuous crop cover across all seasons.

B. Improved Tillage Practices

Adoption of improved tillage methods enhances soil structure by minimizing compaction and erosion, increasing soil carbon sequestration, and maintaining or improving soil fertility. These practices support the accumulation of organic matter and stimulate microbial activity, thereby improving nutrient cycling and leading to more stable and higher crop yields.

- i. Traditionally, most cotton farmers in the Nagpur district performed deep ploughing (over 30 cm) annually, followed by one or two harrowing operations using cultivators and rotavators. These tillage activities typically occurred between the second week of May and the end of the month.
- ii. Farmers associated with ASVATA, within the same geographical area, have adopted regenerative tillage practices. These farmers now conduct shallow tillage (<15 cm) only once every two to three years, followed by one or two harrowing passes during cotton cultivation. Tillage operations are usually carried out from the end of April to mid-June. On average, primary tillage depth ranges from 10–15 cm, while secondary tillage is between 6–8 cm. Additionally, approximately 30–40% of these farmers now use bullock-drawn implements for field preparation.

C. Reduced Use of Synthetic Fertilizers

RPG farmers have fully transitioned away from inorganic fertilizers in cotton cultivation, relying entirely on organic inputs. Most organic farmers utilize Farm Yard Manure (FYM), vermicompost, Jeevamrutha, biofertilizers, and green manures. Maintaining and gradually improving soil fertility is essential for achieving economically viable yields. To produce organic cotton, it is crucial to enhance and sustain the organic matter content of the soil, thereby improving its physical properties, structure, and nutrient availability. Given the limited availability of large quantities of FYM, combining different sources with diverse biological properties is recommended to meet crop nutrient demands effectively.

D. Crop Residue Management

Conventional farmers typically remove and burn cotton stalks immediately after harvest. In contrast, program farmers leave cotton stalks in the field post-harvest to prevent soil erosion caused by wind and rain. These residues are either incorporated into the soil using rotavators or are uprooted and converted into biochar or compost. The estimated quantity of cotton plant residue is approximately 2.7 tons per hectare.

1.13 Project Location

The project is being executed in the states of Maharashtra, and Telangana as depicted in the map presented below.

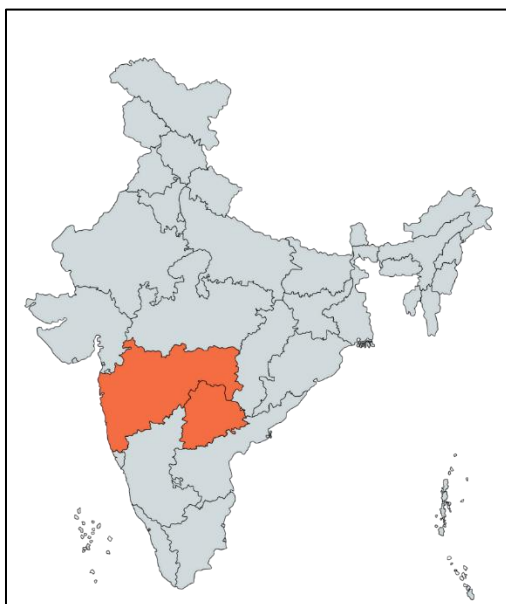


Figure: Project Locations

1.14 Conditions Prior to Project Initiation

- **Ecosystem type:** The ecosystem type before the project initiation was terrestrial ecosystem with land classified as agricultural lands.
- **Current and historical land-use:** The chosen project sites have a historical background of being utilized for agricultural purposes. To validate that the project locations are indeed agricultural or croplands, a Land Use Land Cover (LULC) map will be created for the project area. This map will serve as evidence of the land's agricultural nature.
- **Climatic conditions:** Maharashtra has typical monsoon climate, with hot, rainy and cold weather seasons. Tropical conditions prevail all over the state, and even the hill stations are not that cold. Dew, frost, hail can also be happened sometimes according to the seasonal weather²
 - a. Summer: March, April and May are the hottest months. During April and May thunderstorms are common all over the state. Temperature varies between 22°C-39°C during this season.

² https://www.maharashtratourism.net/climate.html#google_vignette

- b. **Rainy:** Rainfall starts normally in the first week of June. July is the wettest month in Maharashtra, while August too gets substantial rain. Monsoon starts its retreat with the coming of September from the state.
 - c. **Winter:** Cool dry spell, with clear skies gentle breeze and pleasant weather prevails from November to February. But the eastern part of Maharashtra sometimes receives some rainfall. Temperature varies between 12°C-34°C during this season.
 - d. **Rainfall:** Rainfall in Maharashtra differs from region to region. Thane, Raigad, Ratnagiri and Sindhudurg districts, receive heavy rains of an average of 200 centimeters annually. But the districts of Nasik, Pune, Ahmednagar, Dhule, Jalgaon, Satara, Sangli, Solapur and parts of Kolhapur get rainfall less than 50 centimeters. Rainfall particularly concentrates to the Konkan and Sahyadrian Maharashtra. Central Maharashtra receives less rainfall. However, under the influence of the Bay of Bengal, eastern Vidarbha receives good rainfall in July, August and September.
- **Hydrology:** In central India, groundwater is a primary source of drinking water and supports more than 70–80% of the water demand for agriculture. Groundwater resources are heavily dependent on the Indian summer monsoon (June–September), as 90% of the annual rainfall receives only during the monsoon season and 10% of the annual rainfall in the remaining seasons of the year. Groundwater resources in the northern region of central India become more vulnerable due to a deficit in monsoon rainfall³.
 - **Soil type and Health:** India exhibits a diverse range of soil types, reflecting its varied geography and climate. The most prevalent soil types appear to be alluvial soils, black soils, and red soils, which are found in significant quantities across multiple states. Alluvial soils are particularly dominant in the northern plains, with Uttar Pradesh having the largest area (23,089.58 Ha) covered by alluvial soils. Other states with substantial alluvial soil coverage include Bihar, Madhya Pradesh, Punjab, and West Bengal. This is consistent with the presence of major river systems like the Ganges and its tributaries in these regions. Black soils, often associated with cotton-growing regions, are most prominent in Maharashtra (15,423.78 Ha) and Madhya Pradesh (15,203.57 Ha). These soils are also found in significant amounts in Gujarat, Karnataka, and Andhra Pradesh, indicating their prevalence in central and western India. Red soils cover large areas in several states, with Madhya Pradesh (16,601.28 Ha) and Andhra Pradesh (9,930.11 Ha) having the most extensive coverage. These

³ <https://www.mdpi.com/2306-5338/9/12/222#:~:text=In%20central%20India%2C%20groundwater%20is,extraction%20%5B23%2C28%5D>.

soils are also common in Karnataka, Tamil Nadu, and Odisha, suggesting their prevalence in peninsular India⁴.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

India does not regulate GHG emissions from agriculture in the smallholder farming systems. By adhering to following laws and regulations, the project will demonstrate its commitment to promoting sustainable agriculture practices while ensuring compliance with local, regional, and national legal frameworks in the project area

- Fertilizer Control Order, 1985
- Seeds Bill, 2004
- Insecticide (Amendment) Act, 2000
- Protection of Plant Varieties and Farmers' Rights Act, 2001
- Farmers & Farm Workers Commission Act, 2017
- Environment (Protection) Act, 1986
- Air (Prevention and Control of Pollution) Act, 1981
- Water (Prevention & Control of Pollution) Act, 1974
- National Biodiversity Act, 2002: Compliance with this act is ensured by promoting regenerative practices that conserve biodiversity by minimizes agrochemical use hence increase soil diversity.
- Air (Prevention and Control of Pollution) Act, 1981: Compliance with this act will be ensured by implementing practices that reduce air pollution associated with agricultural activities, such as promoting the avoidance of burning crop residues.
- Water (Prevention & Control of Pollution) Act, 1974: The project's regenerative practices, including fertigation and reduced tillage, prevent water pollution and promote water conservation. It ensures compliance with regulations for water quality protection by minimizing nutrient runoff and soil erosion.
- Environment (Protection) Act, 1986: The project ensures compliance with this act by implementing environmentally sustainable practices and minimizing adverse impacts on the environment. This includes adhering to regulations related to air and water quality waste management, and environmental pollution. The project promotes

⁴ <https://360analytika.com/state-wise-area-by-major-soil-type-in-india/>

regenerative practices to reduce environmental pollution and conserve natural resources.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

If yes, provide the registration number and the date of project inactivity under the other GHG program.

Is the project active under the other program?

☐ Yes ☒ No

Project proponents, or their authorized representative, must attest that the project is no longer active in the other GHG program in the Registration Representation.

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

If yes, provide the program name(s), reason(s) and date for the rejection, justification of eligibility under the VCS Program, and any other relevant information.

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes

☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes

☒ No

Supply Chain (Scope 3) Emissions

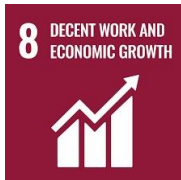
Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes

☒ No

1.18 Sustainable Development Contributions

1.18.1 Sustainable Development Contributions Activity Description

SDG	SDG Impact
	The adoption of advanced regenerative agricultural practices promotes increased crop productivity and economic well-being, empowering individuals to achieve a sustainable livelihood. Through the use of effective methods, both the quality and quantity of agricultural output are enhanced, helping to ensure adequate food availability for all.
	This project aligns with the SDG target 4.3 by providing vocational trainings to the individuals associated with the project activity. By engaging individuals from local communities, the initiative promotes equal access to education and technical skills needed for effective participation in ALM activities.
	Unsustainable agricultural practices which affect the productivity of land, which in turn affect the quality and the quantity of the produce. Due to this, a large section of farmers is heavily affected, and it is more pronounced for the small and marginal farmers. Often the only source of income for these farmers/landowners is agriculture, making them vulnerable to poverty in case of crop failure. This implementation of project activities will create job opportunities and support regional development by providing training and employment in rural areas.

	The project activity involves the implementation of ALM activities over an area of 2,200 ha across the project boundary. This plantation activity will help in the reduction/removal of net GHG emissions from the atmosphere and the addition of more oxygen into the surrounding environment, thus creating a clean environment. An estimated total 225,533 tCO₂e will be removed during a crediting period of 30 years. Therefore, the project activity will promote climate mitigation and adaptation strategies.
	The ALM activities will promote various ecosystem services including improvement of the microclimate of the area, conservation of soil by reducing soil erosion, enhancing ground water quality & quantity, and increasing soil health of the project area.

NITI Aayog is the central institution responsible for driving the attainment of SDGs in India, spearheading the 2030 Agenda through a collaborative and competitive approach among states. The organization works rigorously to monitor all the 17 Goals over 100 indicators nationwide.

1.18.2 Sustainable Development Contributions Activity Monitoring

Table 1: Sustainable Development Contributions

Row number	SDG target	SDG indicator	Net impact on SDG indicator	Current project contributions	Contributions over project lifetime
1)	2.4	2.4.1 Proportion of agricultural area under productive and sustainable development.	Implemented activities to increase	Implemented ALM activities on 5,000 ha for MP01	The project aims to promote sustainable agricultural activities in more than 20,000 ha of area during the project lifetime.
2)	4.3	Project-specific indicator: Number of individuals who received any vocational training to enable their employment in project activity or elsewhere.	Implemented activities to decrease	Vocational training sessions with respect to implementation of sustainable agricultural practices.	Contribute to increasing vocational and relevant skills of local individuals by providing non-formal education and training on issues related to climate change, implementation of project activities and its management, with specific skill-building in operations and surveying activities related to ALM and its monitoring under VCS

3)	8.3	Project Specific Indicator: Number of local individuals employed during the project lifetime.	Implemented activities to increase	A total of 5 individuals were directly employed in Maharashtra and Telangana, respectively during the MP01	Contribute to the generation of employment in the informal sector by contracting locals with target employment of 15 employees for varying lengths of time over the project lifetime. Employment comprises all persons of working age who, during a short reference period (minimum one week), or full time (more than 6 months) will be engaged in any activity to produce goods or provide services for pay or profit.
4)	13.0	Tonnes of greenhouse gas emissions avoided or removed	Implemented activities to increase	Total carbon sequestration in the current MP is 119,583 tCO ₂ e	Prevented the release of 19,51,250 tonnes of carbon into the atmosphere
5)	15.3	15.3.1 Proportion of land that is degraded over total land area	Implemented activities to decrease	The total area covered under the current MP is 5,000	The project activity aims to restore 20,000 area of land during the project lifetime.

1.19 Additional Information Relevant to the Project

Leakage Management

The aim of the project is to increase organic input from farmyard manure, compost to increase soil nutrient and soil carbon. The crop residue produced after harvest is integrated into the soil.

The possible source of leakage could be increase of fuel wood and grazing activity outside the project boundary due to decrease in supply for fodder and fuel for cooking. However, the cropland in the project area do not allow grazing in any time of the year. Therefore, there was no displacement of pre-project grazing activities.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description

Further Information

No further information is added.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification

A three-step approach is adopted by ASVATA for identifying the relevant stakeholders associated with the project throughout its lifetime. The following figure represents the approach adopted for stakeholder identification:

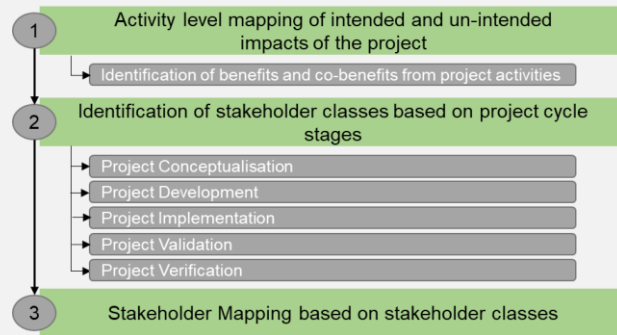


Figure 1: Stakeholder's identification process

The first step involves developing a logical framework model for the designed project to identify activities based on intended and un-intended impacts associated with the project activities. This step included the identification of benefits and co-benefits aligned with the project activity to identify the potential stakeholders associated at different stages of the project cycle. A multi-method approach combining desk review, field scoping, and participatory mapping to identify stakeholders under the three-step approach. The following table provides a description of the different approaches adopted to identify the stakeholders.

S. No.	Activity	Description
1	Desk Review	Review land use records, historical LULC data, development plans, and existing stakeholder data
2	Key Informant Consultations	Interview officials, elders, IP leaders, and civil society organizations to identify actors not captured in records
3	Participatory Community Mapping	Facilitate site-level community mapping to identify areas used for farming, rituals, settlements etc.
4	Categorize Stakeholders	Classify stakeholders into: a) Direct stakeholders (e.g., farmers, local residents); and b) Indirect stakeholders (e.g. NGOs, media, academic institutions etc.)
5	Profile and Analyze	For each stakeholder group, determine: location, role, access/tenure status, vulnerability, cultural or economic importance, and engagement preference.

To identify the stakeholders, a list of potential types/categories of stakeholders was created. The stakeholder's types/categories identified were as follows:

- Community/neighbour
- Landowner/ farmers
- Local policymaker
- NGOs
- Employees
- National association/National focal point
- Companies
- Universities/research institutes

Using this list as a guide, a list of stakeholders to be proactively contacted was created per country. The list of stakeholders included all landowners and farmers

	that were participating or expected to participate in the Project at that time
Legal or customary tenure/access rights	The farmers participating in the Project are either themselves the landowners of the land they are farming on or hold a lease agreement to this land.
Stakeholder diversity and changes over time	The local stakeholder group comprises farmers and landowners within the project area where local consultations are conducted. Below is a delineation of the diversity within each stakeholder category along with potential shifts in their composition over time. • Small holder farmers: Predominantly rural-based, these stakeholders are deeply entrenched in agricultural activities, forming the backbone of the local economy. Their livelihoods depend on the performance of crop and livestock markets, rendering their income subject to fluctuations influenced by climatic conditions, market trends, and governmental policies. The lifestyle of farmers and landowners varies significantly based on factors such as farm size, agricultural practices adopted, and access to resources. These disparities shape their perspectives and priorities concerning the project. • Community members: The community comprises individuals with strong ties to rural livelihoods, underscoring their affinity for agrarian activities and traditional practices. They exhibit a mosaic of social, economic, and cultural backgrounds, contributing to the vibrancy and heterogeneity of the local population. Economic diversity is evident within the community, with disparities in income levels, employment opportunities, and access to resources shaping individual experiences and aspirations. Cultural norms, traditions, and values prevalent within the community influence stakeholder interactions, perceptions of the project, and decision-making processes. • Local Policymakers: The local policymakers have authority and influence in shaping agricultural policies and developmental initiatives. Their work often revolves around agricultural issues, positioning them as crucial intermediaries between governmental bodies and grassroots stakeholders. • Non-Governmental Organizations (NGOs): NGO employees typically specialize in areas related to agriculture, championing causes aligned with sustainable farming practices, rural development, or

	environmental conservation. These stakeholders are driven by a shared commitment to effecting positive change within the agricultural landscape. NGOs play a pivotal role in amplifying the voices of marginalized stakeholders, providing technical expertise, and facilitating community engagement initiatives. • Employees: They are employed by the PP, These stakeholders contribute to the operational and logistical facets of agricultural endeavors. They plays an important role in connecting stakeholders to the project and are important for stakeholder engagement. • Universities/Research Institutes: This additional stakeholder category encompasses individuals engaged in academia or pursuing education within institute/university settings. Their focus on scientific research related to agriculture brings a specialized perspective to the stakeholder landscape. Interactions among these diverse stakeholder groups further enrich the dialogue surrounding the project's objectives and implementation strategies. These stakeholder groups are expected to stay made up of a similar make-up in the future.
Expected changes in well-being	The stakeholders most likely to be impacted by the project are smallholder's farmers of the project area. As provided in the section 1.12, it is expected that the project relative to baseline scenario will have positive impact on the environment such as: • Increase carbon sequestration in the soil and therewith improved soil quality. • Reduced erosion and improved soil quality. • Increased resilience to extreme weather events such as droughts and flooding. • Improved water retention in the soil, less runoff and • Reducing pollution of water bodies creating improved water quality. • Improved biodiversity such as increased development of micro-organisms, strengthening of ecosystems. • Reduced pollutants in the soil, air and water, due to the reduced use of chemical fertilisers Moreover, the project provides the following co benefits such as: • Knowledge and skill enhancement: Farmers will gain awareness of sustainable agricultural methods

	through training sessions conducted by the project partners (PP), fostering local skill development. • Income boost: Farmer livelihoods will improve as they earn additional income through carbon credits. • Encouragement for young farmers: The project aims to motivate the younger generation to remain in or engage with farming activities. • Potential for both direct and indirect employment opportunities on farms.
Location of stakeholders	All identified stakeholder groups, as outlined in the table above, were actively encouraged to participate in the consultation process conducted within the project areas of Maharashtra and Telangana. It is anticipated that local communities, and customary rights holders will not be affected by the project.
Location of resources	The project will encompass project activity instances location in the states of Maharashtra and Telangana.

2.1.2 Stakeholder Consultation and Ongoing Communication

Date of stakeholder consultation	30-April-2025
Stakeholder engagement process	The aim of the stakeholder consultation is to meaningfully engage local stakeholders and to discuss environmental, social, and economic impacts, both positive contributions and potential risks that the project may have during planning, implementation and operational stages of the project. During these stakeholder consultation meetings, a diverse mix of participants attended, including farmers, landowners, government employees, researchers, NGOs and ASVATA employees.
Consultation outcome	During the stakeholder consultation meeting, the following questions were discussed: • Farmer eligibility for the carbon project • Cropping systems and common agricultural practices • Awareness regarding carbon credits • Queries regarding third-party auditing • Discussion on benefit-sharing mechanisms

	- Questions about the monitoring process for adopted farming practices - Discussion on the social and environmental impact of project All questions, discussions, and responses are documented comprehensively in the meeting minutes and feedback forms.
Ongoing communication	Ongoing conversations are kept open, via email, WhatsApp, and phone calls in addition to in-person meetings where applicable, such that project stakeholders can always reach a project proponent representative
Stakeholder input	All valuable discussions and inputs from stakeholders who participated in the consultation meetings were carefully documented and integrated into various sections of this PD. Common feedback, such as inquiries regarding training and guidance on carbon farming, as well as local agronomic advice, was raised by the farmers. In response, the project is actively addressing these concerns by organizing continuous training sessions. Additionally, regular awareness meetings are being conducted at the village level to emphasize the significance of regenerative agriculture and highlight carbon as a co-benefit, encouraging farmers to adopt these practices.



2.1.3 Free Prior and Informed Consent

Obtaining consent	The principle of Free, Prior, and Informed Consent (FPIC) ensures that farmers have the right to self-determination and to give or withhold consent to projects affecting their ancestral domains, lands, territories, and resources. It is a fundamental process for recognizing farmer's rights, building trust, and legitimizing project presence in areas with cultural, spiritual, and territorial significance. FPIC was secured at the initiation of the carbon project development by the farmer communities within the project area. Comprehensive details about the planned project activities, involvement of farmers, expected outcomes, benefits and co-benefits and potential risks associated with the project were provided by the PP to the communities. FPIC survey forms were signed by representatives of the communities.
Outcome of FPIC	Apart from the farmers, there are no customary right holders of the land nor are there any indigenous people or other local community members who hold the authority to determine or consent to the activities conducted on the land that is part of this Project.

2.1.4 Grievance Redress Procedure

Development process	The feedback and grievance redressal mechanism has the following objectives: • To inform decision making related to project design and development, integrating feedback into the project. • To serve as a mechanism for timely resolution of a problem and prevent problems from escalating into social conflict. • To be part of the project monitoring and evaluation process and contribute to learnings.
Grievance redress procedure	The project proponent ensures that a structured mechanism is in place to address stakeholder disputes during project planning and implementation. A complaint register is maintained at the project office for farmers to record their concerns and all registered grievances are reviewed and resolved in a timely manner. Farmers can also directly contact the PP team via phone for support

during project implementation, and any concerns raised are recorded and addressed promptly.

2.1.5 Public Comments

Public comments are to be addressed in the Final PD.

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

The project proponent (ASVATA) have relevant experience managing sustainable agriculture activities.

2.2.2 Risk Assessment

	Risks identified	Mitigation or preventative measure taken
Natural and human-induced risks to stakeholders' wellbeing	No risk identified	Project proponent continuously ensures that stakeholders' wellbeing is protected from any natural and human-induced risks
Risks to stakeholder participation	No risk identified.	Stakeholders have the freedom to choose whether or not they wish to participate in the project.
Working conditions	No risk identified	Compared to the baseline scenario, the working conditions on the farms will remain similar. The project activities such as reducing tillage, improving crop residue management, avoidance of residue burning and fertigation will result in no significant change in working conditions.
Safety of women and girls	No risk identified	Compared to the baseline scenario, the Project will have no impact on the safety of minority and marginalized groups, including children.
Safety of minority and marginalized groups, including children	No risk identified	Compared to the baseline scenario, the Project will have no impact on the safety of minority and marginalized groups, including children.
Pollutants (air, noise, discharges to water, generation of waste, and	No risk identified	The project will decrease the pollutants, notably by reducing the use of high-input fertilizers through fertigation. This shift is expected to

release of hazardous materials and chemical pesticides and fertilizers)		have a positive environmental impact, as the projected farming practices are predicted to enhance water retention in the soil, thereby diminishing surface runoff. Monitoring of fertilizer usage will be conducted throughout the project, with any changes in fertilizer application carefully documented and accounted.
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2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

Risks identified ⁵		Mitigation or preventative measure(s) taken
Discrimination	No risk identified	The project has a zero-tolerance policy toward discrimination and all employees are made aware of this upon hiring, as per the Indian labour law.
Sexual harassment	No risk identified	The project proponent has a zero-tolerance policy toward sexual harassment
Gender equity in labor and work	No risk identified	The Project is open to all farmers no matter their gender. All farmers receive the same terms and conditions in the formal agreement. All farmers must comply with local, regional and national laws, including all laws regarding gender equity and equal pay for labour and work.
Forced labor Child labor Human trafficking	No risk identified	The project strictly complies with Indian laws and regulations pertaining to human trafficking, forced labor, and child labor, including but not limited to the Immoral Traffic (Prevention) Act, the Bonded Labour System (Abolition) Act, the Child Labour (Prohibition and Regulation) Act, and the Juvenile Justice (Care and Protection of Children) Act.

⁵ The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

2.3.2 Human Rights

Not applicable for this Project.

The Project area does not have indigenous peoples or alternative customary rights holders other than those involved in the Project. The local communities were invited to participate in the local stakeholder consultation process

2.3.3 Indigenous Peoples and Cultural Heritage

The Project area does not have indigenous people or cultural heritage that is at risk of being lost due to the project activities and thus does not require protection.

2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken
Rights to territories and resources	The ownership of the land solely belongs to the farmers who have adopted regenerative practices and are enrolled in this carbon project.

2.3.5 Benefit Sharing

Process used to design the benefit sharing plan	The benefit-sharing plan is designed to allocate a significant portion of the revenue from carbon credit sales to participant farmers, enhancing their livelihoods and improving the surrounding environment. The remaining revenue is designated to cover project development and management costs for Asvata.
Summary of the benefit sharing plan	The Landowner will receive incentives in accordance with the mutually agreed terms, contingent upon the Project Proponent's successful generation of carbon revenue following the project's validation and registration under the Verified Carbon Standard (VCS).
Approval and dissemination of benefit sharing plan	The benefit sharing mechanism has been agreed upon after due consultations among the project participants and approved by the project management. The benefit sharing plan is duly mentioned in the agreement made among the participant farmer, and ASVATA, and disseminated to the farmers participated in

	this project, during the consultations and stakeholder meetings conducted by the project team and took the approval of the beneficiaries through signing of the agreement.
Benefit sharing during the monitoring period	A Monitoring Committee comprising representatives of ASVATA and RPGF will ensure that the share of the benefits from the sale of the Carbon Credits are passed on to the participating farmers.

2.4 Ecosystem Health

	Risks identified	Mitigation or preventative measure(s) taken
Impacts on biodiversity and ecosystems	No risk identified	The project activities will positively impact biodiversity and ecosystems, such as increased development of micro-organisms thereby strengthening ecosystems
Soil degradation and soil erosion	No risk identified	The project activities will reduce soil degradation and soil erosion
Water consumption and stress	No risk identified	Water consumption and stress is expected to remain the same as in the baseline of the project.

2.4.1 Rare, Threatened, and Endangered species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☐ Yes ☒ No

2.4.2 Introduction of species

Not applicable as the project does not involve the introduction of any invasive species into the project area. The crop varieties utilized by farmers are regionally accepted, and the project activities strictly adhere to this principle. Neither do any of the project activities allow any invasive species to thrive through project implementation. The different crop varieties reported by farmers in the project area will be duly recorded, particularly for cotton.

2.4.3 Ecosystem conversion

The Project does not clear or drain existing natural ecosystems and has not done so in at least the 10 years prior to the project start date. For each project instance, a GIS check will be done to confirm that the project area has not been cleared of natural ecosystems in the past 10 years and is not located on a wetland, peatland, or forest.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	VM0042	Methodology for improved agricultural land management	2.0
Tool	VCS	AFOLU Non-Permanence Risk Tool	2.0 (Online version)

3.2 Applicability of Methodology

Methodology ID	Applicability condition	Justification of compliance
VM0042	Project must introduce or implement one or more new changes to pre-existing agricultural management practices; Reduced (organic or inorganic) application, improved water management, residue incorporation, improved crop planting and harvesting, and improve grazing practices.	Condition met. The project aims to implement multiple activities like reduced inorganic application of fertilizers, residue management, crop rotation and improved tillage practices. Details are given in section 1.12.
...	Any quantitative adjustment (e.g., decrease in fertilizer application rate) must exceed 5% of the pre-existing value.	Condition met. The quantitative adjustments in project scenario will be more than 5% of baseline scenario.
	The project activities must be implemented on either cropland or grassland at the project start date and remain cropland or grassland throughout the project crediting period	Condition met. The project involves only croplands as per the methodological guidelines.
	The project area must not have been cleared of Native ecosystems within the last ten years prior to the project start date	Condition met. LULC is done to prove the applicability condition.

	The project activity proposed is not expected to result in a sustained reduction of greater than 5% in productivity.	The project activities are implemented to increase the productivity of the crops.
	If the project activity involves the application of biochar, it must be produced using feedstock that would otherwise have been left to decay in aerobic or anaerobic conditions or been burned in an uncontrolled manner.	Not applicable.
	The project activities occur on a wetland; however, this condition does not exclude crops subject to artificial flooding, provided it is demonstrated that crop cultivation does not impact the hydrology of any nearby wetlands.	The project area is not falling under the category of wetlands.

3.3 Project Boundary

Source		Gas	Included?	Justification/Explanation
Baseline	Soil Organic Carbon	CO ₂	Yes	The proposed project activities will increase the overall net SOC pool in the project scenario.
		CH ₄	No	Excluded as the project activity dose no cause significant change
		N ₂ O	No	Excluded as the project activity dose no cause significant change
		Other	No	Excluded as the project activity dose no cause significant change
	Use of nitrogen fertilizers	CO ₂	No	Excluded as the project activity dose no cause significant change
		CH ₄	No	Excluded as the project activity dose no cause significant change
		N ₂ O	Yes	The project activities encourage the reduction of N fertilizer by either site-specific nutrient

Source		Gas	Included?	Justification/Explanation
Project				management or by replacing it with organic fertilisers.
		Other	No	Excluded as the project activity dose no cause significant change
	Soil Organic Carbon	CO ₂	Yes	The proposed project activities will increase the overall net SOC pool in the project scenario.
		CH ₄	No	Excluded as the project activity dose no cause significant change
		N ₂ O	No	Excluded as the project activity dose no cause significant change
		Other	No	Excluded as the project activity dose no cause significant change
	Use of nitrogen fertilizers	CO ₂	No	Excluded as the project activity dose no cause significant change
		CH ₄	No	Excluded as the project activity dose no cause significant change
		N ₂ O	Yes	The project activities encourage the reduction of N fertilizer by either site-specific nutrient management or by replacing it with organic fertilisers.
		Other	No	Excluded as the project activity dose no cause significant change

3.4 Baseline Scenario

As per VM0042 v2.0 methodology, continuation of pre-project agricultural management practices is the most plausible baseline scenario. The conventional agriculture practices in baseline scenario of the project area includes practices such as:

1. Excessive fertiliser application through broadcasting
2. Burning of crop residue
3. Intensive tillage practice

The baseline scenario is determined for each sample unit (e.g. for each field) by applying a historical look-back period to produce an annual schedule of activities to be repeated over the first crediting period. This is determined via the collection of information pertaining to reduced fertilizer use, water management, reduced tillage, crop residue management and avoidance of burning practices during the three-year period immediately before the project start date. Information relating to agricultural management activities was sourced via farmer signed attestation, survey and remote sensing analysis.

3.5 Additionality

3.5.1 Regulatory Surplus

Is the project registered or seeking registration in an UNFCCC Annex 1 or Non-Annex 1 country?

- ☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

- ☐ Yes ☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

- ☐ Yes ☒ No

If no, describe which mandated laws, statutes, or other regulatory frameworks require project activities and provide evidence of systematic non-enforcement to demonstrate regulatory surplus.

3.5.2 Additionality Methods

As per the latest version of the VCS Standard, section 3.14.1 and methodology VM0042 v2.1 requirements, the additionality of this project is demonstrated in two steps:

1. The project shall demonstrate regulatory surplus at validation and each project crediting period renewal.
 - a. Regularity surplus
 - b. Barrier analysis
 - c. Common practice analysis
2. Additionality shall be demonstrated and assessed in accordance with the requirements set out in the methodology applied to the project.

3.6 Methodology Deviations

Not applicable. There is no methodology deviation.

4 IMPLEMENTATION STATUS

4.1 Implementation Status of the Project Activity

For all projects, describe the implementation status of the project activity(s), including information on the following:

- *The operation of the project activity(s) during the monitoring period, including any information on events that may impact the GHG emission reductions or removals and monitoring.*
- *Any other changes (e.g., to project proponent or other entities).*

For AFOLU projects, include information on the following:

- *Where no new project activities have been implemented during the current monitoring period, demonstrate that previously implemented project activities continued to be implemented during the current monitoring period.*
- *Report any loss of carbon stock that occurred during the monitoring period. The date of the loss(es), date of discovery(s), size (hectares impacted) and extent (tCO₂e) of the loss must be described. Specify if the loss meets the definition of a loss event and/or reversal. In all cases, justify how the project meets VCS requirements related to loss events and reversals.*

5 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

Describe the procedure for quantification of baseline emissions and/or carbon stock changes in accordance with the applied methodology. Baseline emissions may be negative where carbon stock increases (sinks) exceed baseline emissions. Specify the reductions and removals separately where the applied methodology provides procedures and equations to do so. Include all relevant equations here and provide sufficient information to allow the reader to reproduce the calculations. Explain and justify all relevant methodological choices (e.g., with respect to selection of emission factors and default values). Include all calculations in the emission reduction and removal calculation spreadsheet.

5.2 Project Emissions

Describe the procedure for quantification of project emissions and/or carbon stock changes in accordance with the applied methodology. Project emissions may be negative where carbon stock increases (sinks) exceed project emissions. Specify the reductions and removals separately where the applied methodology provides procedures and equations to do so. Include all relevant equations here and provide sufficient information to allow the reader to reproduce the calculations. Explain and justify all relevant methodological choices (e.g., with respect to selection of emission factors and default values). Include all calculations in the emission reduction and removal calculation spreadsheet.

5.3 Leakage Emissions

Describe the procedure for quantification of leakage emissions in accordance with the applied methodology. Specify the reductions and removals separately where the applied methodology provides procedures and equations to do so. Include all relevant equations here and provide sufficient information to allow the reader to reproduce the calculations. Explain and justify all relevant methodological choices (e.g., with respect to selection of emission factors and default values). Include all calculations in the emission reduction and removal calculation spreadsheet.

5.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

Describe the procedure for the quantification of estimated GHG emission reductions (reductions) and carbon dioxide removals (removals). Include all relevant equations.

For data and parameters monitored, use the estimated data/parameter values provided in Section 6.2 below. Document how each equation is applied, in a manner that enables the reader to reproduce the calculations. Provide calculations for all key equations to allow the reader to reproduce the annual calculations for estimated reductions or removals. Specify the reductions and removals separately where the applied methodology provides procedures and equations to do so. Include all of the above in the emission reduction and removal calculation spreadsheet.

Complete the tables below by vintage period (calendar year). Note that the baseline or project emissions subtotals may be negative where sinks exceed emissions. Only specify the estimated VCU for reductions and removals separately where the applied methodology provides procedures and equations to do so.

For projects that are not required to assess permanence risk, complete the table below for the project crediting period.

Vintage period	Estimated baseline emissions (tCO _{2e})	Estimated project emissions (tCO _{2e})	Estimated leakage emissions (tCO _{2e})	Estimated reduction VCUs (tCO _{2e})	Estimated removal VCUs (tCO _{2e})	Estimated total VCUs (tCO _{2e})
DD-MMM-YYYY to 31-Dec-YYYY	<i>Example:</i> 50,000	<i>Example:</i> 20,000	<i>Example:</i> 10,000	<i>Example:</i> 10,000	<i>Example:</i> 10,000	<i>Example:</i> 20,000
01-Jan-YYYY to 31-Dec-YYYY						
01-Jan-YYYY to DD-MMM-YYYY						
Total						

For projects required to assess permanence risk, complete the table below for the project crediting period.

Note that the buffer pool allocation is split proportionally between the estimated reductions and removals. (For example, if a project is estimated to achieve 20,000 tCO_{2e} removals and 80,000 tCO_{2e} reductions and has a buffer contribution of 20%, or 20,000, the estimated removal VCUs would be 16,000 and reduction VCUs would be 64,000).

Vintage period	Estimated baseline emissions (tCO _{2e})	Estimated project emissions (tCO _{2e})	Estimated leakage emissions (tCO _{2e})	Estimated buffer pool allocation (tCO _{2e})	Estimated reduction VCUs (tCO _{2e})	Estimated removal VCUs (tCO _{2e})	Estimated total VCU issuance (tCO _{2e})
DD-MMM-YYYY to 31-Dec-YYYY	<i>Example:</i> 50,000	<i>Example:</i> 20,000	<i>Example:</i> 10,000	<i>Example:</i> 4,000	<i>Example:</i> 8,000	<i>Example:</i> 8,000	<i>Example:</i> 16,000
01-Jan-YYYY to 31-Dec-YYYY							
01-Jan-YYYY to DD-MMM-YYYY							
Total							

6 MONITORING

6.1 Data and Parameters Available at Validation

Data / Parameter	
Data unit	Indicate the unit of measure
Description	Provide a brief description of the data/parameter
Source of data	Indicate the source(s) of data
Value applied:	Provide the value applied
Justification of choice of data or description of measurement methods and procedures applied	Justify the choice of data source, providing references where applicable. Where values are based on measurement, include a description of the measurement methods and procedures applied (e.g., what standards or protocols have been followed), indicate the responsible person/entity that undertook the measurement, the date of the measurement and the measurement results. More detailed information may be provided in an appendix.
Purpose of data	Indicate one of the following: - Determination of baseline scenario (AFOLU projects only) - Calculation of baseline emissions - Calculation of project emissions - Calculation of leakage
Comments	Provide any additional comments

6.2 Data and Parameters Monitored

Data / Parameter	
Data unit	Indicate the unit of measure
Description	Provide a brief description of the data/parameter
Source of data	Indicate the source(s) of data

Description of measurement methods and procedures applied	<i>Specify the measurement methods and procedures, any standards or protocols followed, and the person/entity responsible for the measurement. Include any relevant information regarding the accuracy of the measurements (e.g., accuracy associated with meter equipment or laboratory tests).</i>
Frequency of monitoring/recording	<i>Specify measurement and recording frequency</i>
Value applied:	<i>Provide an estimated value for the data/parameter</i>
Monitoring equipment	<i>Identify equipment used to monitor the data/parameter including type, accuracy class, and serial number of equipment, as appropriate.</i>
QA/QC procedures applied	<i>Describe the quality assurance and quality control (QA/QC) procedures applied, including the calibration procedures where applicable.</i>
Purpose of data	<i>Indicate one of the following:</i> <i>Calculation of baseline emissions</i> <i>Calculation of project emissions</i> <i>Calculation of leakage</i>
Calculation method	<i>Where relevant, provide the calculation method, including any equations, used to establish the data/parameter.</i>
Comments	<i>Provide any additional comments</i>

6.3 Monitoring Plan

Describe the process and schedule for obtaining, compiling, and analyzing the monitored data and parameters set out in Section 6.2 (Data and Parameters Monitored) above.

Include details on the following:

- The methods used for measuring, recording, storing, aggregating, collating and reporting on monitored data and parameters. Where relevant, include the processes used for calibrating monitoring equipment.*
- The organizational structure, responsibilities and competencies of the personnel that carried out monitoring activities.*
- The procedures for internal auditing and QA/QC.*
- The procedures for handling non-conformances with the validated monitoring plan.*

- Any sampling approaches used, including target precision levels, sample sizes, sample site locations, stratification, frequency of measurement and QA/QC procedures.

Where appropriate, include line diagrams to display the GHG data collection and management system.

7 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

7.1 Data and Parameters Monitored

Data / Parameter	
Data unit	Indicate the unit of measure
Description	Provide a brief description of the data/parameter
Value applied:	Provide the monitored value for the data/parameter
Comments	Provide any additional comments

7.2 Baseline Emissions

Quantify the baseline emissions and/or carbon stock changes for the monitoring period in accordance with the applied methodology. Baseline emissions may be negative where carbon stock increases (sinks) exceed baseline emissions. Specify the reductions and removals separately where the applied methodology provides procedures and equations to do so. Include all relevant equations here and provide sufficient information to allow the reader to reproduce the calculation. Include all calculations in the emission reduction and removal calculation spreadsheet.

7.3 Project Emissions

Quantify project emission and/or carbon stock changes for the monitoring period in accordance with the applied methodology. Project emissions may be negative where carbon stock increases (sinks) exceed project emissions. Specify the reductions and removals separately where the applied methodology provides procedures and equations to do so. Include all relevant equations here and provide sufficient information to allow the reader to reproduce the calculation. Include all calculations in the emission reduction and removal calculation spreadsheet.

7.4 Leakage Emissions

Quantify leakage emissions for the monitoring period in accordance with the applied methodology. Specify the reductions and removals separately where the applied methodology provides procedures and equations to do so. Include all relevant equations here and provide sufficient information to allow the reader to reproduce the calculation. Include all calculations in the emission reduction and removal calculation spreadsheet.

7.5 GHG Emission Reductions and Carbon Dioxide Removals

Quantify the GHG emission reductions (reductions) and carbon dioxide removals (removals) for the monitoring period. Include all relevant equations.

Complete the tables below by vintage period (calendar year). Note that the baseline or project emissions subtotals may be negative where sinks exceed emissions. Only specify the estimated VCUs for reductions and removals separately where the applied methodology provides procedures and equations to do so.

For projects that are not required to assess permanence risk, complete the table below for the project crediting period:

Vintage period	Baseline emissions (tCO ₂ e)	Project emissions (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Reduction VCUs (tCO ₂ e)	Removal VCUs (tCO ₂ e)	Total VCUs (tCO ₂ e)
DD-MMM-YYYY to 31-Dec-YYYY	Example: 50,000	Example: 20,000	Example: 10,000	Example: 10,000	Example: 10,000	Example: 20,000
01-Jan-YYYY to 31-Dec-YYYY						
01-Jan-YYYY to DD-MMM-YYYY						
...						
Total						

For projects required to assess permanence risk:

i) Provide the requested information using the table below:

State the non-permanence risk rating (%)

Has the non-permanence risk report been attached as either an appendix or a separate document?	☐ Yes ☐ No
For ARR and IFM projects with harvesting, state, in tCO ₂ e, the Long-term Average (LTA).	
Has the LTA been updated based on monitored data, if applicable?	☐ Yes ☐ No If no, provide justification.
State, in tCO ₂ e, the expected total GHG benefit to date	
If a loss occurred (including a loss event or reversal), state the amount of tCO ₂ e lost:	

Vintage period	Baseline emissions (tCO ₂ e)	Project emissions (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Buffer pool allocation (tCO ₂ e)	Reductions VCU's (tCO ₂ e)	Removals VCU's (tCO ₂ e)	Total VCU issuance (tCO ₂ e)
DD-MMM-YYYY to 31-Dec-YYYY	Example: 50,000	Example: 20,000	Example: 10,000	Example: 4,000	Example: 8,000	Example: 8,000	Example: 16,000
01-Jan-YYYY to 31-Dec-YYYY							
01-Jan-YYYY to DD-MMM-YYYY							
Total							

Vintage period	Ex-ante estimated reductions/removals	Achieved reductions/removals	Percent difference	Explanation for the difference
DD-MMM-YYYY to 31-Dec-YYYY				
01-Jan-YYYY to 31-Dec-YYYY				

...				
01-Jan-YYYY to DD-MMM-YYYY				
Total				

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

Use the table below to describe the commercially sensitive information included in the project description to be excluded in the public version.

Section	Information	Justification

APPENDIX X: <TITLE OF APPENDIX>

Use appendices for supporting information. Delete this appendix (title and instructions) where no appendix is required.